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API TESTER

🔗

Interactive Tester

QCMS Developer API Overview

v1.0 Stable

Welcome to the QCMS (Quality & Continuous Improvement Management System) Integration API. Our API allows external applications (such as Ideation Tools, ERP systems, CRMs, and custom workplace software) to securely sync continuous improvement ideas, Kaizen projects, and quality records into QCMS.

🛡️ Secure Authentication

Bearer token authentication hashed using SHA-256 with per-organization isolation.

⚡ High Velocity

Built-in rate limiting (100 req/min), sliding window protection, and sub-50ms latency.

🏗️ Modular Architecture

Future-proof REST design for seamless expansion to Users, Projects, and Webhooks.

2. Authentication

All requests to the QCMS Integration API must include a valid Organization API key passed as a Bearer token in the **Authorization** HTTP request header.

Copy

Authorization: Bearer qcms_live_83hd92jd93jd83jd93jd93jd

⚠️ Security Best Practices:

- Never expose secret API keys in client-side applications or public code repositories.
- Always transmit requests over HTTPS.
- Regenerate keys periodically or immediately if compromised under **Settings** → **Integrations**.

3. Base URL

The QCMS Integration API is served over HTTPS. Use the following base URLs depending on your environment:

Environment	Base URL
Production	https://api.qcms.com/v1/
Local Development	http://localhost:5000/api/v1/integrations

4. API Key Lifecycle Management

Organization Administrators can manage API keys in QCMS via **Settings** → **Integrations**:

- Generate API Key:** Creates a new live secret key (format `qcms_live_...`).
- Regenerate Key:** Invalidates the current key and issues a new key immediately.
- Enable/Disable Key:** Temporarily pause API traffic without deleting credentials.
- Delete Key:** Permanently revokes access.

5. Quick Start Guide

- 1

Generate API Key in QCMS Platform (Settings → Integrations).
- 2

Copy Secret Token (e.g. `qcms_live_TvvFkQ6zRRI6YIAUAA4CL...`).

- 3

Make HTTP POST Request to `/api/v1/integrations/ideas` with JSON body.
- 4

Receive **201 Created** response confirming idea storage.

6. Endpoint: Import Approved Idea

POST `/api/v1/integrations/ideas`

Pushes an approved continuous improvement idea from external tools (Ideation Tool, ERP) into the QCMS database.

Request Body Fields

FIELD NAME	TYPE	REQUIRED	DESCRIPTION
<code>ideaCode</code>	string	REQUIRED	Unique identifier code for the idea (e.g., <code>IDA-2026-006</code>). Used for duplicate checking.
<code>title</code>	string	REQUIRED	Short title describing the continuous improvement initiative.
<code>presentSituation</code>	string	OPTIONAL	Current problem statement or defect situation description.
<code>proposedSolution</code>	string	OPTIONAL	Proposed corrective action or solution.
<code>category</code>	string	OPTIONAL	Category (Quality, Cost, Delivery, Environment, Morale, Safety). Default: <code>Quality</code> .
<code>department</code>	string	OPTIONAL	Target department name (e.g. <code>Production</code>).
<code>submittedBy</code>	string	OPTIONAL	Primary submitter full name.
<code>coSuggesters</code>	array[string]	OPTIONAL	List of co-submitter names (e.g. [<code>"Smith"</code> , <code>"David"</code>]).
<code>tangibleBenefit</code>	number	OPTIONAL	Estimated financial savings in currency.
<code>intangibleBenefit</code>	string	OPTIONAL	Qualitative benefit description (e.g. "Customer Satisfaction").
<code>investmentRequired</code>	number	OPTIONAL	Required capital expenditure.
<code>implementationTime</code>	string	OPTIONAL	Estimated implementation duration (e.g. <code>8 Weeks</code>).
<code>impactLevel</code>	string	OPTIONAL	Impact level (<code>Low</code> , <code>Medium</code> , <code>High</code>). Default: <code>Medium</code> .

7 & 8. Sample Request and Success Response

JSON Request Payload

```
{
  "ideaCode": "IDA-2026-006",
  "title": "Reduce Paint Defects",
  "presentSituation": "Paint rejection increased from 5% to 12% over the last quarter.",
  "proposedSolution": "Install automatic viscosity monitoring and adjust spray gun settings.",
  "department": "Production",
  "category": "Quality",
  "submittedBy": "John Doe",
  "coSuggesters": ["David", "Smith"],
  "tangibleBenefit": 100000,
  "intangibleBenefit": "Customer Satisfaction",
  "investmentRequired": 250000,
  "implementationTime": "8 Weeks",
  "impactLevel": "Medium",
}
```

201 Created Response

```
{
  "success": true,
  "message": "Idea imported successfully.",
  "ideaCode": "IDA-2026-006"
}
```

```
"status": "Approved"
}
```

9. Error Codes & Specifications

The API uses standard HTTP response status codes to indicate request outcome:

401 Unauthorized

Missing, Invalid, or Disabled API Key

```
{"error": "Unauthorized", "message": "Invalid or disabled API Key."}
```

409 Conflict

Duplicate Idea Code

```
{"message": "Idea already imported."}
```

429 Rate Limit

Rate Limit Exceeded (100 req/min)

```
{"error": "Too Many Requests", "message": "Rate limit exceeded. Maximum 100 requests per minute allowed."}
```

14. Rate Limits

To ensure system stability, requests are limited to **100 requests per minute** per organization. Exceeding this limit triggers an immediate **429 Too Many Requests** HTTP response.

▶ Interactive API Playground

Test QCMS API requests live in your browser

Send Request

ORGANIZATION API KEY

qcms_live_demo

ENDPOINT

POST /ideas (Import Idea) ▼

REQUEST JSON BODY

```
{
  "ideaCode": "IDA-2026-PLAYGROUND",
  "title": "Automated Quality Inspection System",
  "presentSituation": "Manual inspection error rate at 3.5%",
  "proposedSolution": "Deploy AI Vision sensors on assembly line",
  "department": "Quality Assurance",
  "category": "Quality",
  "submittedBy": "Developer Tester",
}
```